

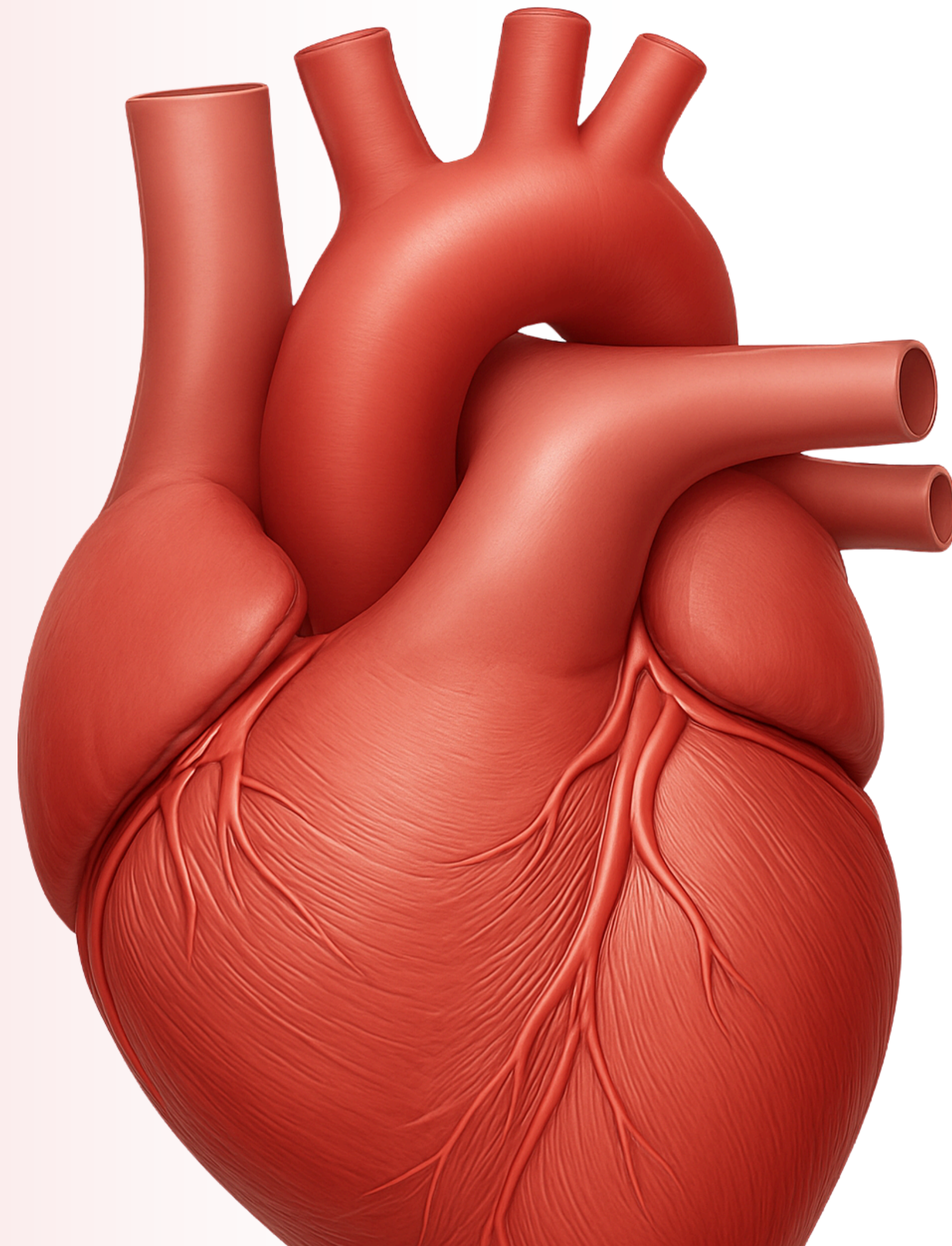
Modeling Cardiac Muscle Fibers in the Left Ventricle and Electrophysiology Simulations

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Course: Cardiac Modelling

Professor: Simone Pezzuto



Outline

1. Introduction
2. Rule-based methods for ventricular fiber generation
3. Numerical results
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 - Idealized ventricular mesh
 - Realistic anatomical ventricle
4. Cardiac Electrophysiology:
Anisotropic and isotropic comparison
5. Conclusions



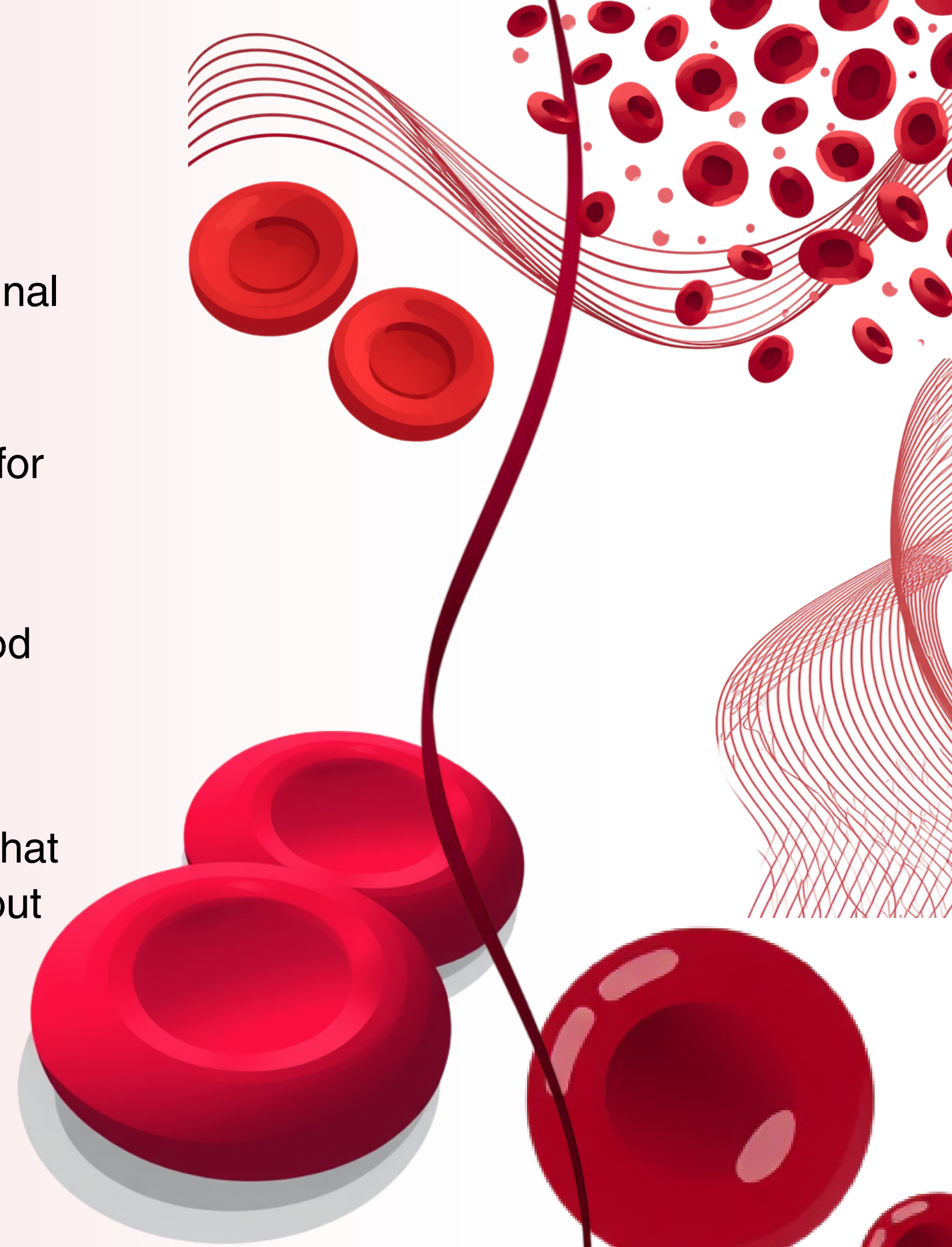
Introduction

The Problem: Cardiac tissue is anisotropic, the electrical signal travels significantly faster along fibers.

Tissue property: incorporating myocardial fibers is needed for more accurate electrophysiology modeling.

Objective: Implement a Laplace-Dirichlet Rule-Based Method (LDRBM) to generate a realistic ventricular fiber field

What are Rule-Based Models? Computational algorithms that assign spatial fiber orientations across the myocardium without requiring patient-specific imaging data

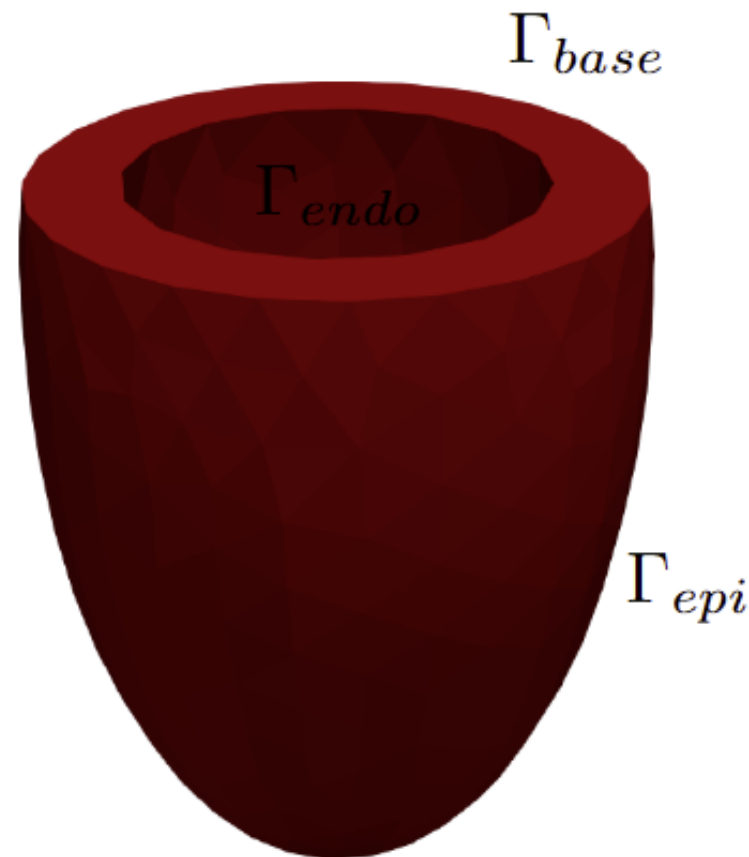


Rossi-Rule-Based Method: Theoretical Workflow

1 Anatomical definition of the domain

$$\Omega_{myo} \subset R^3$$

$$\partial\Omega_{myo} = \Gamma_{epi} \cup \Gamma_{endo} \cup \Gamma_{base}$$



2 Transmural Field

computation of a normalized transmural distance ϕ

$$\begin{cases} -\Delta\phi = 0 & \text{in } \Omega_{myo} \\ \phi = 0 & \text{on } \Gamma_{endo} \\ \phi = 1 & \text{on } \Gamma_{epi} \\ \nabla\phi \cdot \mathbf{n} = 0 & \text{on } \Gamma_{base} \end{cases}$$

Transmural direction vector: $\nu = \nabla\phi$

3 Local basis

outward normal direction to the ventricular base:

$$\mathbf{k} = \mathbf{n}_{base}$$

initial orthonormal coordinate system:

$$P = [\hat{\mathbf{e}}_l, \hat{\mathbf{e}}_n, \hat{\mathbf{e}}_t] = \text{axis}(\mathbf{k}, \nu) = \begin{cases} \hat{\mathbf{e}}_t & = \frac{\nu}{\|\nu\|}, \\ \hat{\mathbf{e}}_n & = \frac{\mathbf{k} - (\mathbf{k} \cdot \hat{\mathbf{e}}_t)\hat{\mathbf{e}}_t}{\|\mathbf{k} - (\mathbf{k} \cdot \hat{\mathbf{e}}_t)\hat{\mathbf{e}}_t\|}, \\ \hat{\mathbf{e}}_l & = \hat{\mathbf{e}}_n \times \hat{\mathbf{e}}_t, \end{cases}$$

Rossi-Rule-Based Models: Theoretical Workflow

4 Rotation of the vectors

- Needed to match the true fiber orientation
- The vectors are rotated to produce a new mapped coordinate system $[f, n, s]$ corresponding to the fiber, cross-fiber and sheet directions.
- The rotational angles vary linearly as a function of the normalized transmural distance:

$$\alpha(\phi) = \alpha_{epi}(1 - \phi) + \alpha_{endo}\phi$$

Where

$$\alpha_{endo} = -60^\circ \quad \alpha_{epi} = +60^\circ$$

Once the transmural distance and the angle are computed, the cardiac fiber field can be constructed

5 Local fiber orientation

$$\mathbf{f}(\phi) = \cos(\alpha(\phi)) \hat{\mathbf{e}}_l + \sin(\alpha(\phi)) \hat{\mathbf{e}}_n$$

This formulation guarantees a smooth vector field

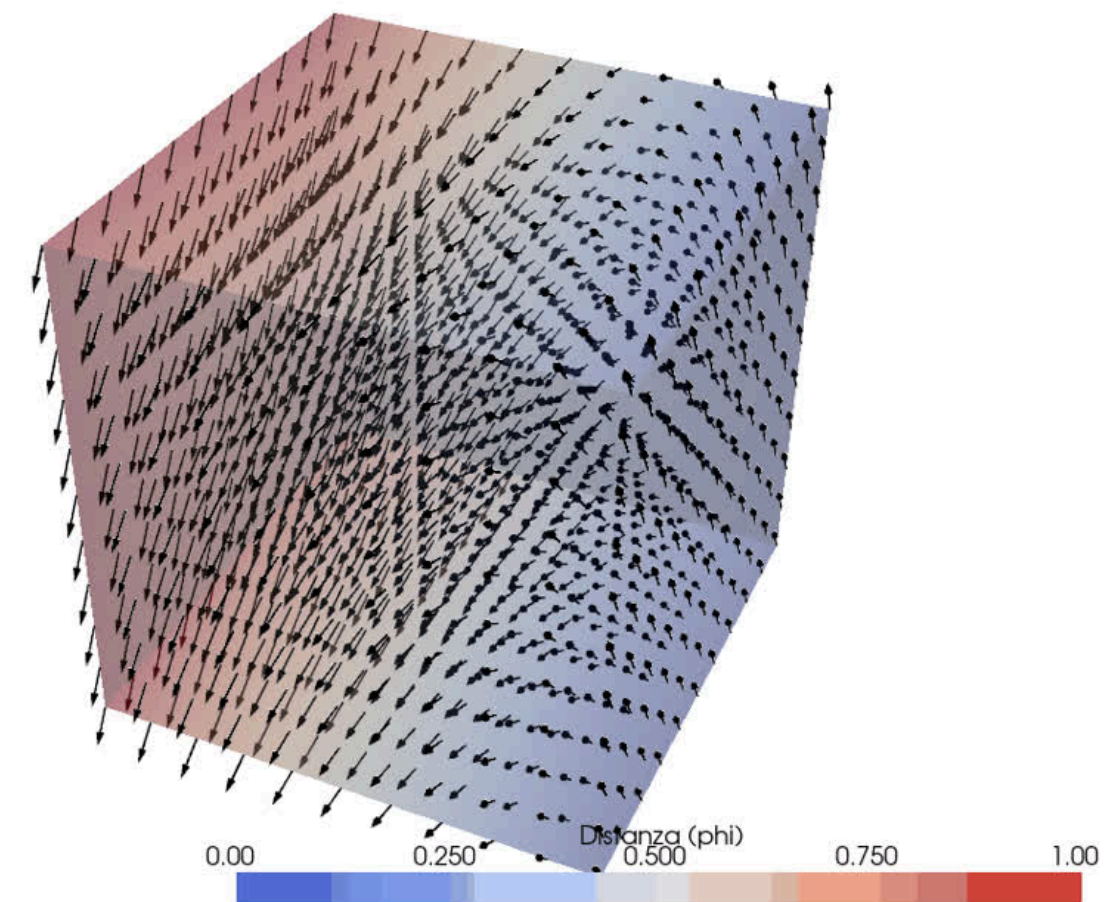
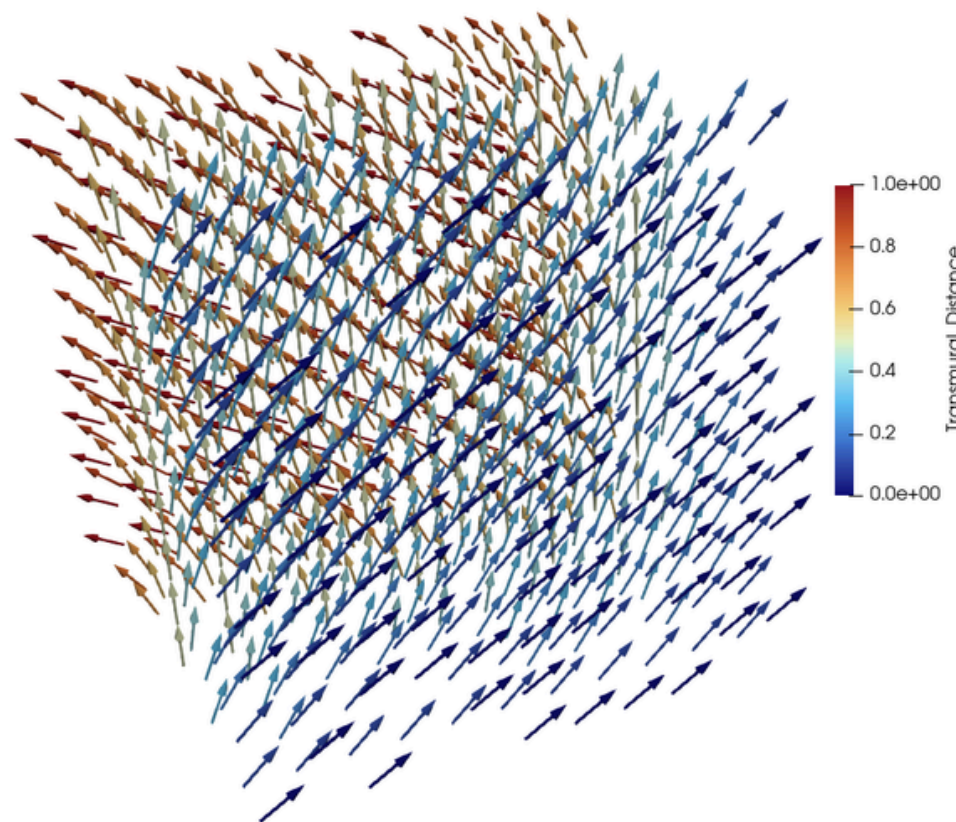
Numerical Results

→ Cubic Domain Validation

Geometry: 3D unit cube $\Omega = [0, 1]^3$

Method: R-RBM implementation in Firedrake

1. Spatial Discretization: 'CG1' finite elements for both scalar and vector function spaces.
2. Weak formulation with Dirichlet BCs
3. Vector field Generation and projection



Outcome:

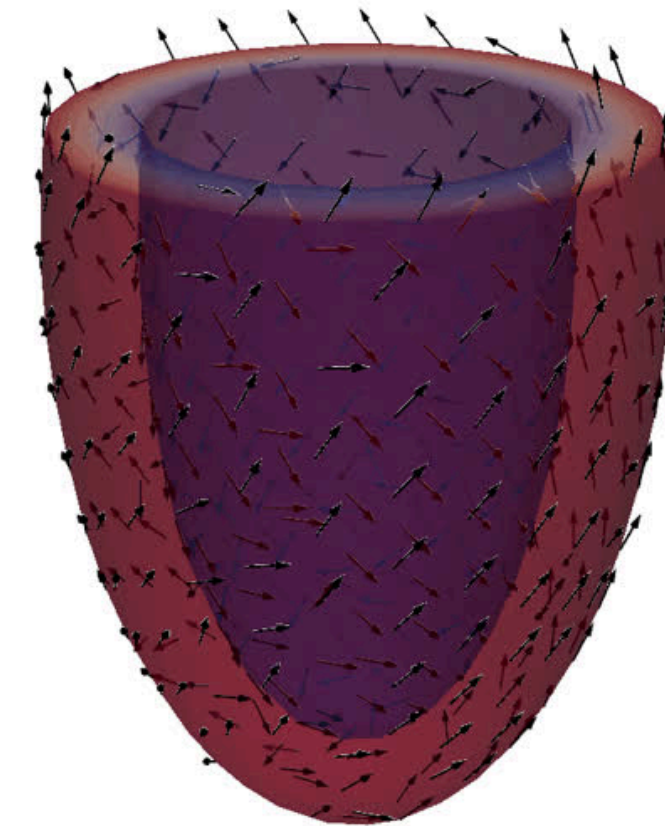
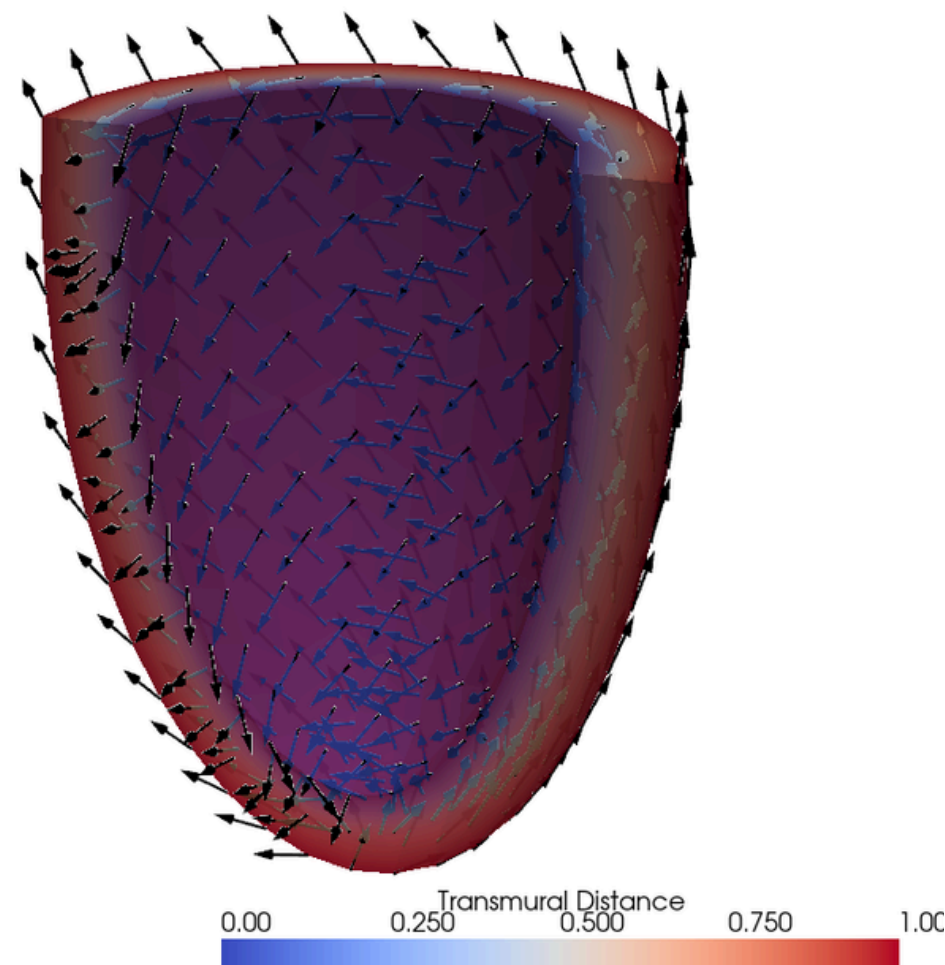
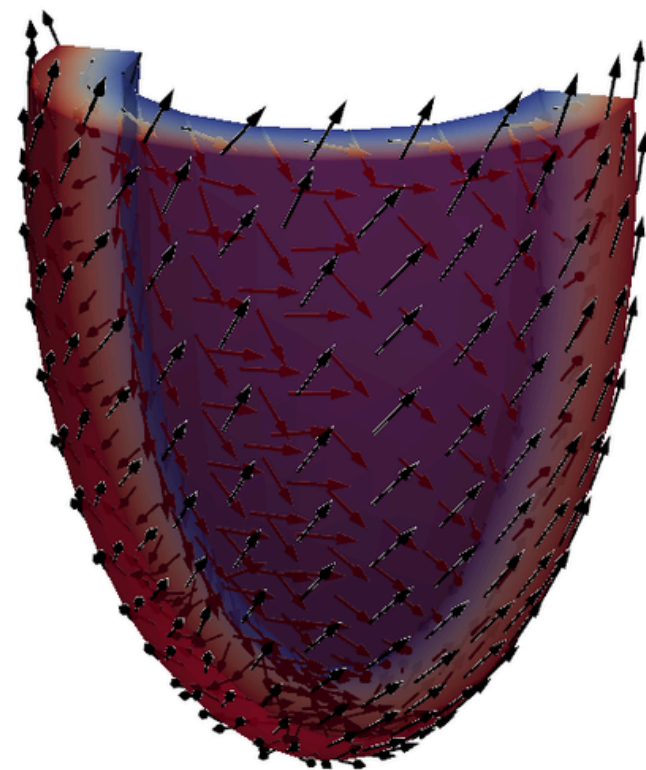
Verified the transmural interpolation of the angle,
Achieved smooth and continuous fiber twist generation.

Numerical Results

→ Idealized Ventricle

Geometry: truncated ellipsoid

Method: R-RBM implementation via Firedrake



Outcome:

- Accurate calculation of the transmural gradient across curvature
- Generation of a physiological double-helical macroscopic structure
- Smooth transmural rotation from endocardium to epicardium

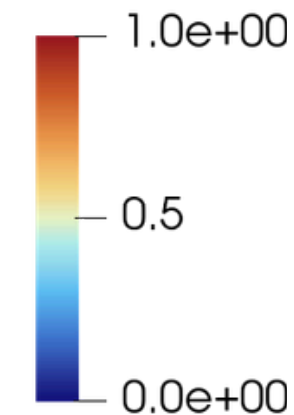
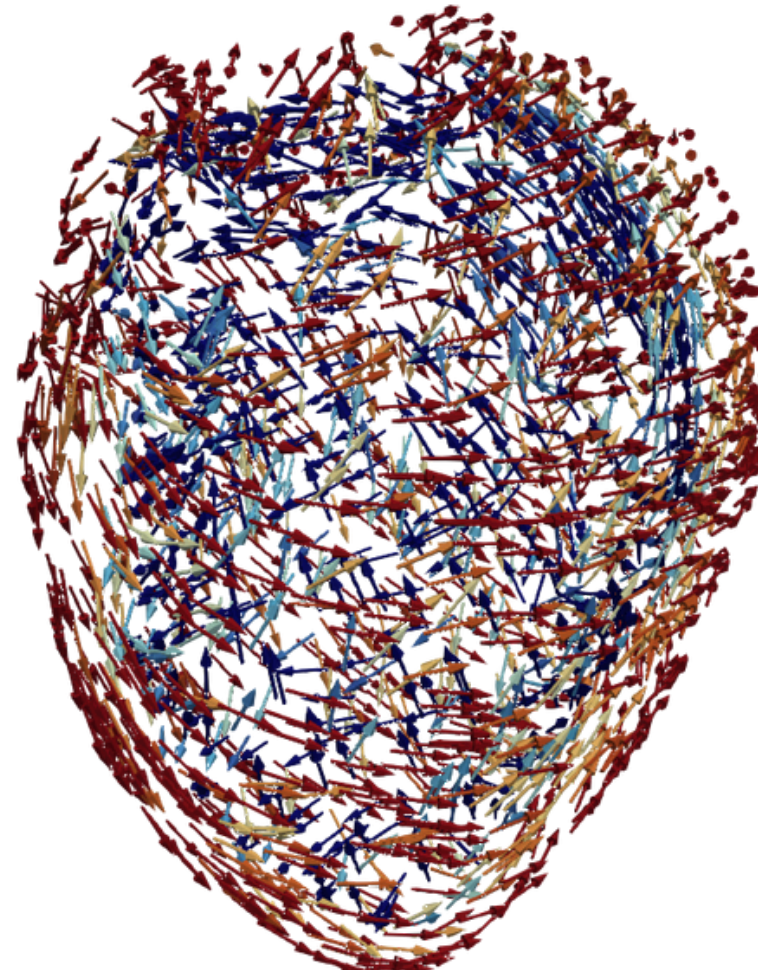
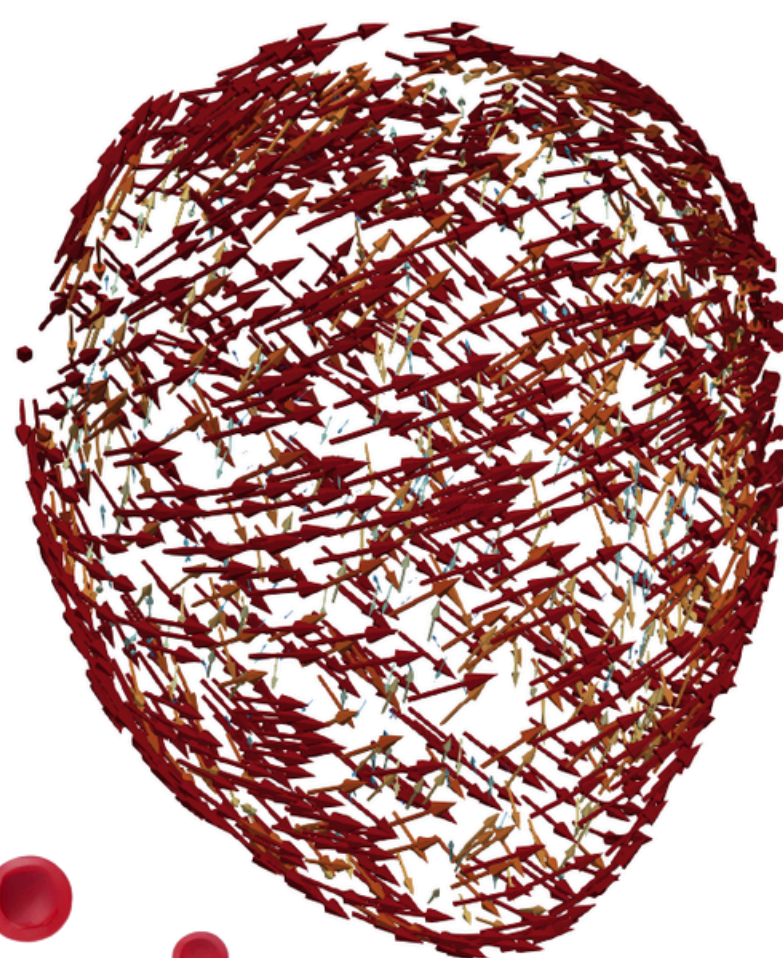
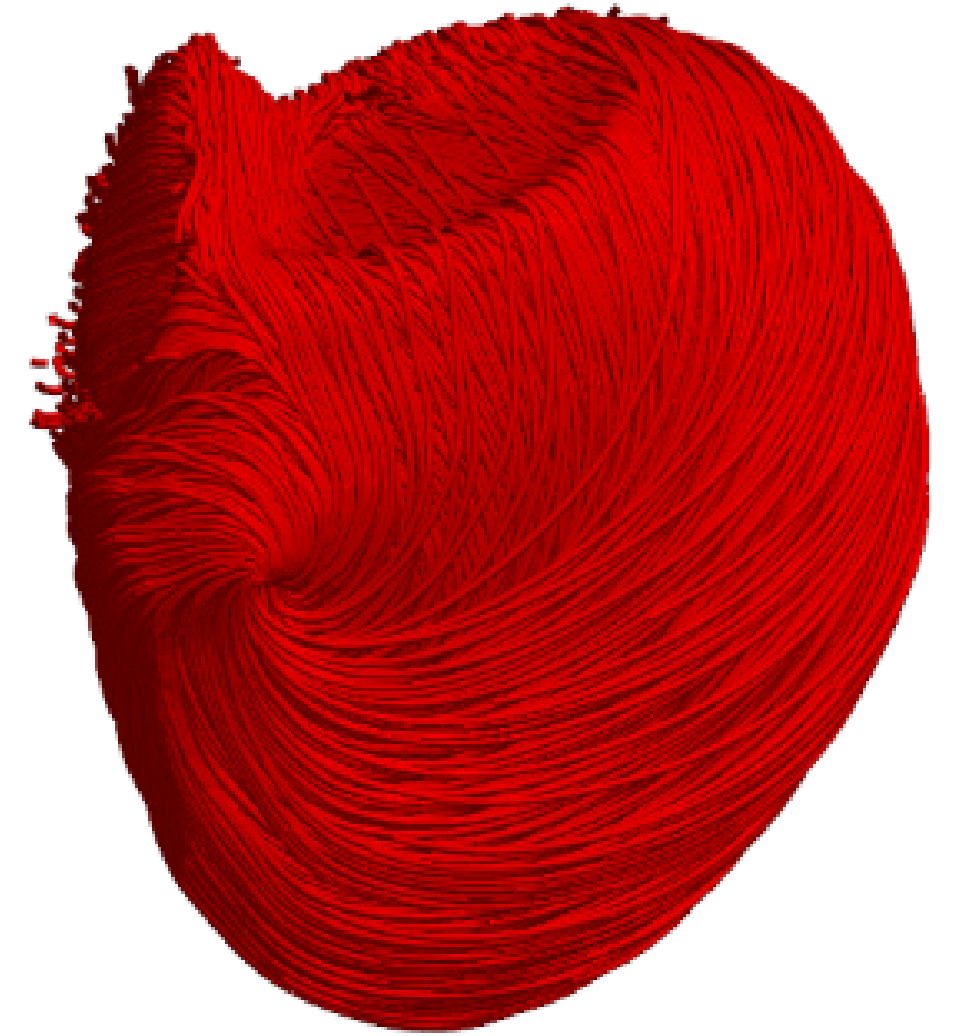
Numerical Results

→ Realistic Anatomical Ventricle

Geometry: Patient-specific left ventricle mesh featuring local variations in wall thickness, irregular curvature and complex boundary regions

Method: R-RBM implementation via Firedrake with unstructured tetrahedral mesh

Outcome: robust generation of the physiological fiber field on a realistic left ventricle anatomy

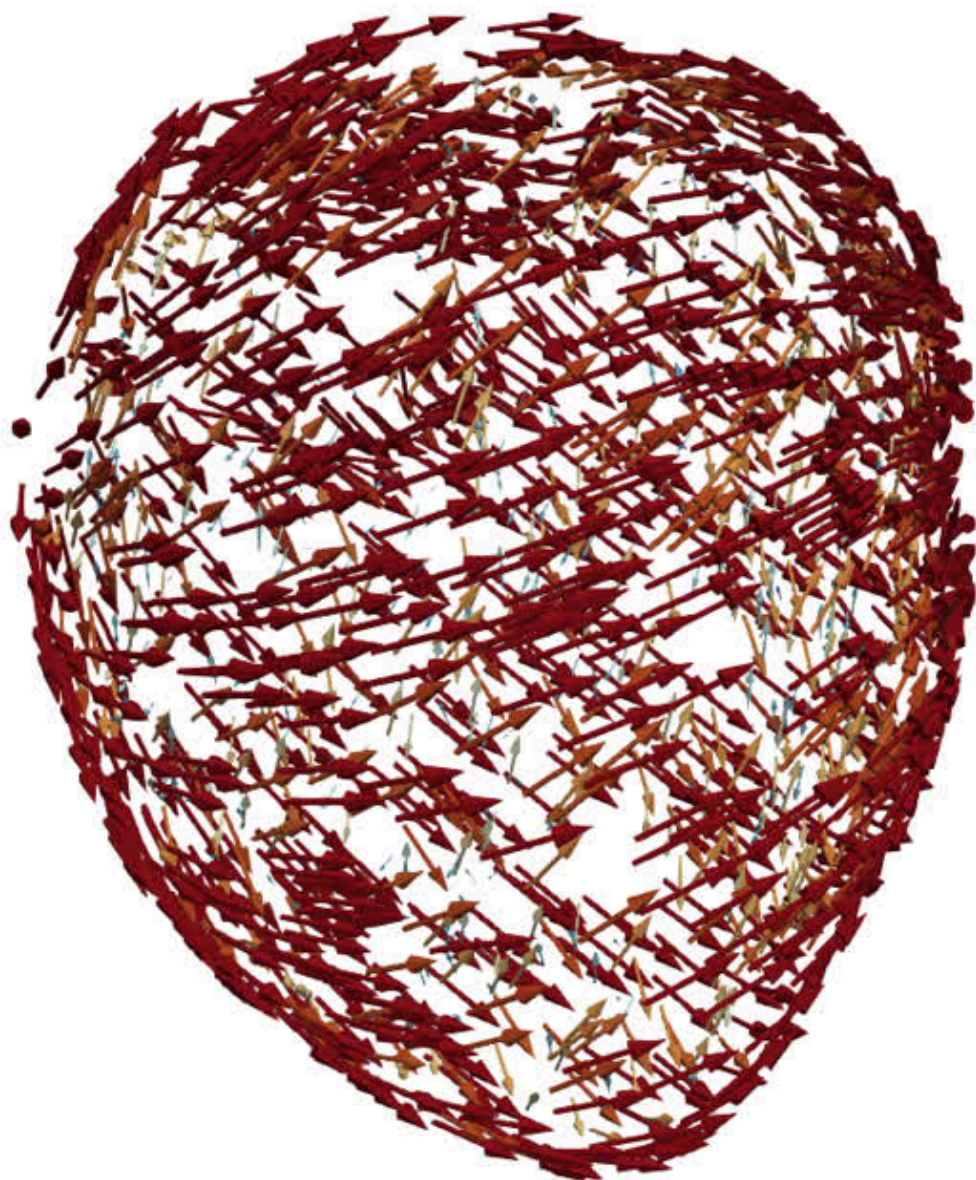


- Smooth transmural tracking from endocardium to epicardium.
- Natural formation of fiber vortices (apical whorl), preserving vector field continuity and boundary conditions.

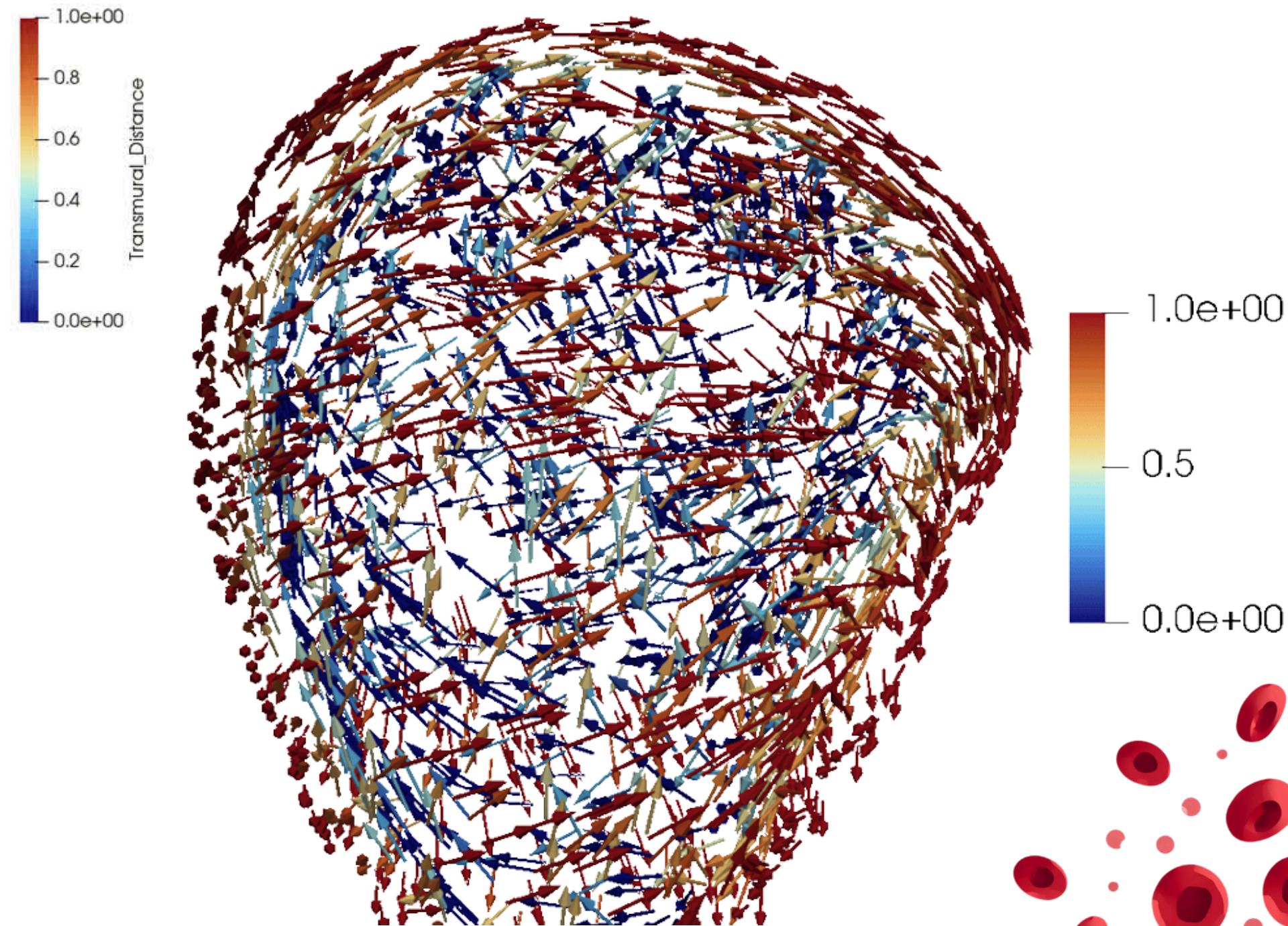
Numerical Results

→ Realistic Anatomical Ventricle

Surface fiber continuity



Transmural fiber continuity



Cardiac Electrophysiology

1 Mathematical model: pure diffusion

$$\frac{\partial V}{\partial t} = \nabla \cdot (\mathbf{D} \nabla V) \quad \text{in } \Omega_{myo} \times (0, T]$$

Neumann Boundary Conditions:

$$(\mathbf{D} \nabla V) \cdot \mathbf{n} = 0 \quad \text{on } \partial \Omega_{myo}$$

Initial Condition:

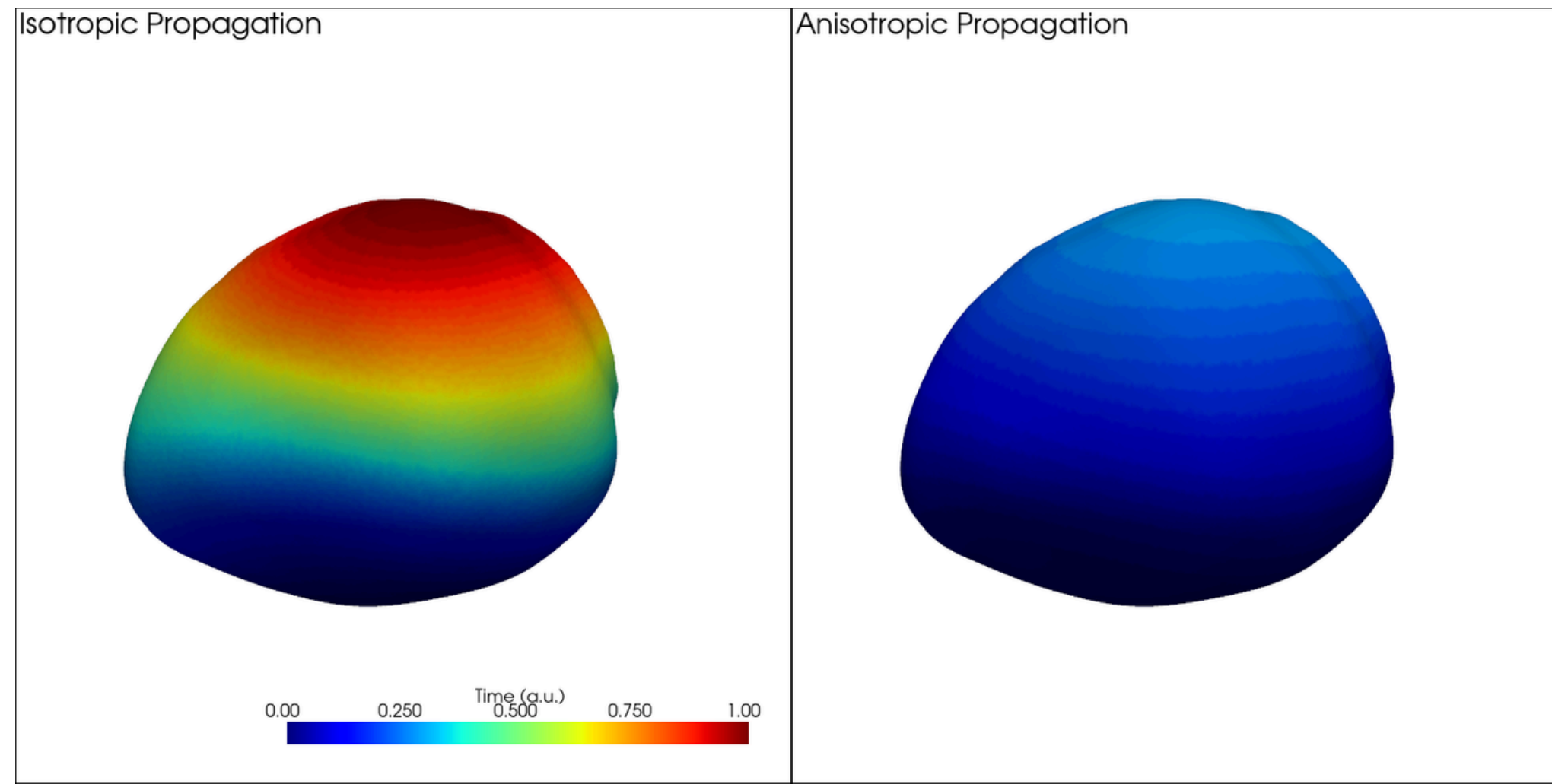
$$V(\mathbf{x}, 0) = \begin{cases} 1.0 & \text{if } \|\mathbf{x} - \mathbf{x}_0\| \leq R \\ 0.0 & \text{otherwise} \end{cases}$$

localized transmural apical stimulus,
R=5mm in simulations.

2 Conductivity tensors comparison:

- *Isotropic tensor:* $\mathbf{D}_{iso} = \sigma_t \mathbf{I}$
- *Anisotropic tensor:* $\mathbf{D}_{ani} = \sigma_t \mathbf{I} + (\sigma_l - \sigma_t)(f \otimes f)$.
transversal anisotropy was assumed

Conductivity ratio: $\sigma_l / \sigma_t \approx 10$

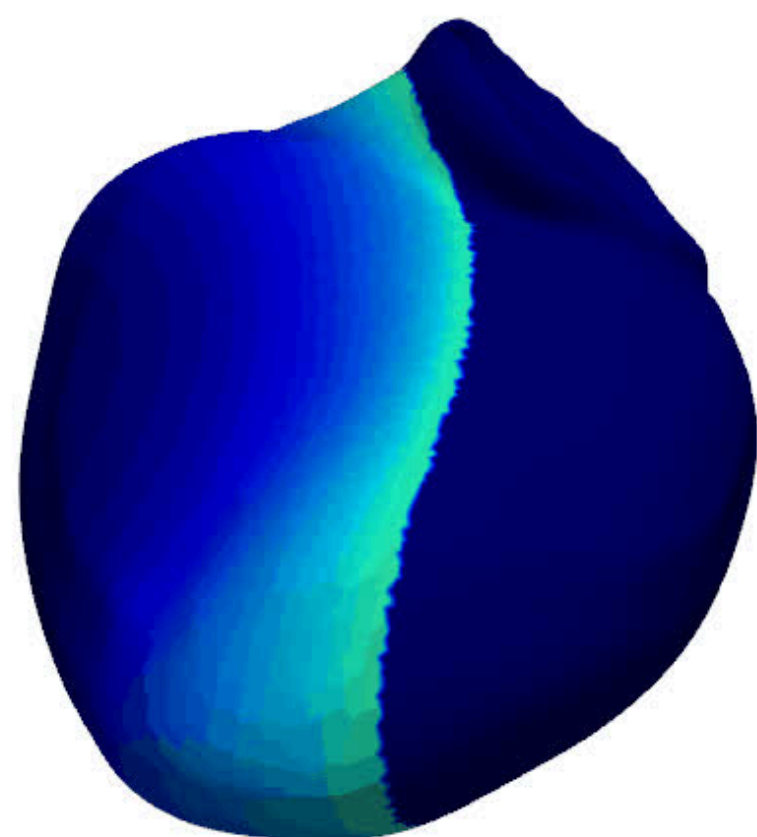


Cardiac Electrophysiology

Isotropic activation map

Spherical wavefront

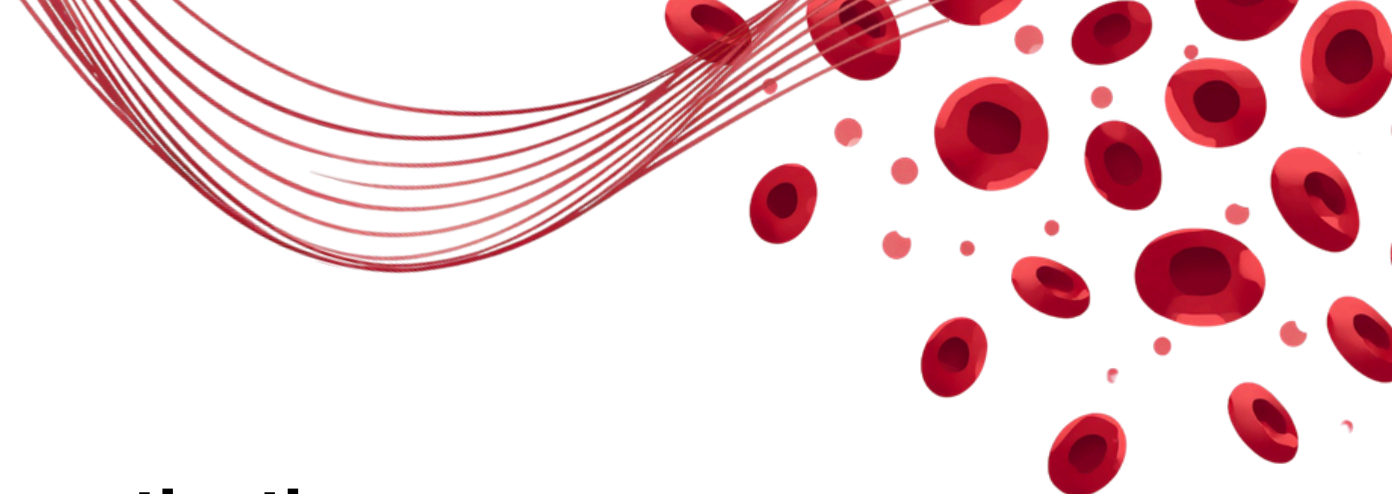
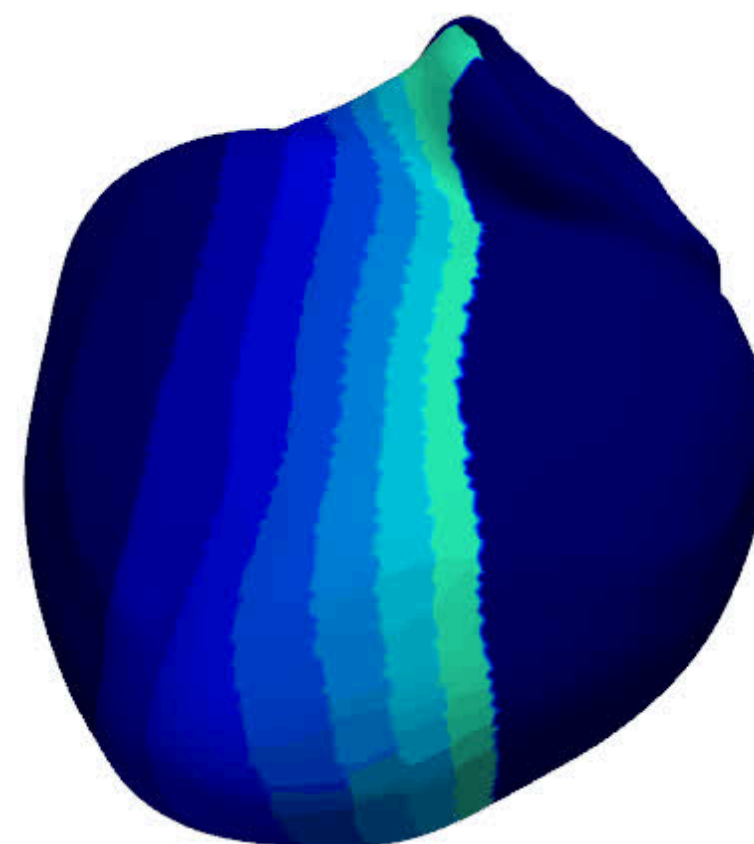
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Anisotropic activation map

Fiber-guided wavefront

Tempo: 34.8 a.u.



Conclusions & Key Takeaways

01

LDRBM Framework:

- *Laplace-Dirichlet Approach:* Successfully implemented to compute smooth transmural distances without experimental imaging (e.g., DTI).
- *Geometric Adaptability:* Validated from simple geometries to complex anatomical left ventricle mesh.

02

Fiber Architecture Generation:

- *Physiological Alignment:* Automated rule-based rotation correctly reproduces the continuous angle variation from endocardium to epicardium.
- *Continuity:* Demonstrated cardiomyocyte orientation wrapping around the entire ventricular volume.

03

Electrophysiological Impact:

- *Isotropic vs. Anisotropic Conduction:* Highlighted the impact of fiber architectures.
- *Anisotropic Conduction:* Successfully shown preferential and higher-velocity electrical propagation directions.